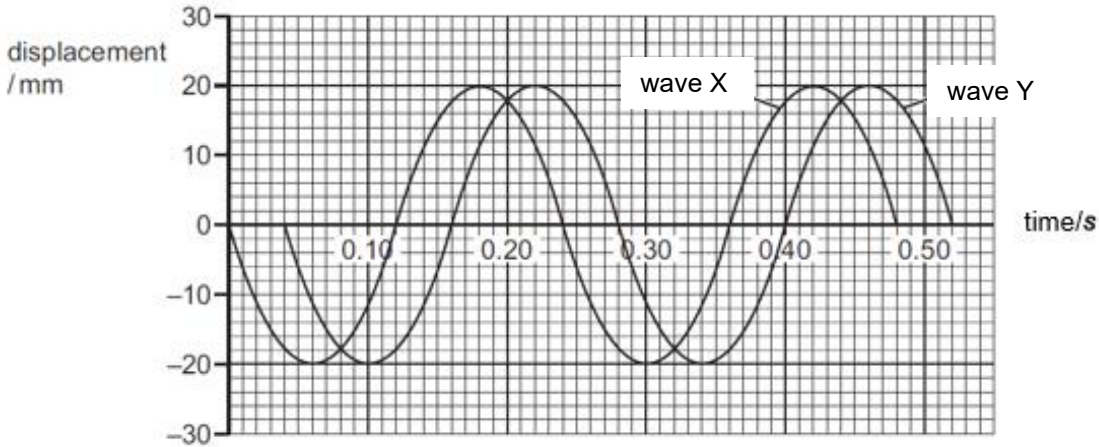
16 The diagram shows the variation with time of the displacement of two transverse progressive waves, X and Y.



Which of the following statements is correct?

A Wave X leads wave Y by $\pi/4$ rad.

B Wave X lags wave Y by $\pi/4$ rad.

C Wave X leads wave Y by $\pi/3$ rad.

D Wave X lags wave Y by $\pi/3$ rad.